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Progress Report

SMARTALYZE: AN AI-ASSISTED DATA VISUALIZATION AND ANALYSIS PLATFORM

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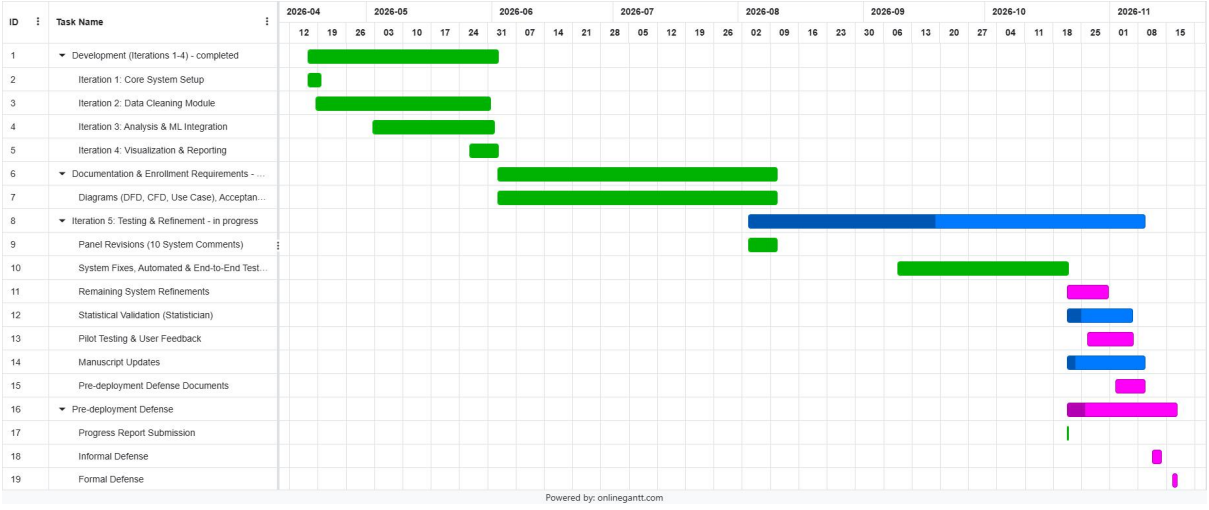
Introduction

Smartalyze is an AI-Assisted web-based platform that enables users with limited technical knowledge, including students, researchers, educators, and small business owners, to upload structured datasets, perform data cleaning, explore and analyze information, and download the processed results. The platform is being developed using the Iterative Software Development Model, which consists of five iterations: Core System Setup, Data Cleaning Module, Analysis and Intelligence Layer, Visualization and Reporting, and System Testing and Refinement.

This report provides an overview of the project's progress during the midterm period. It outlines the activities completed following the outline presentation, the challenges encountered and the solutions implemented to address them, and the project's current progress in relation to the established development schedule. It also identifies the remaining tasks that must be completed in preparation for the pre-deployment defense scheduled for November 11–17, 2026.

Gantt Chart

Figure 1 shows the updated project schedule based on the actual work recorded in the project's version-control history. Green bars are completed tasks, blue bars are tasks in progress (the darker part shows how much is done), and pink bars are upcoming tasks up to the pre-deployment defense. The original plan from the capstone manuscript (Figure 3) is compared with the actual dates in Table 3.



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Figure 1. Updated Gantt chart: actual progress and remaining work (as of October 21, 2026)

Project Status

Overview of the Project's Current Stage

The project has reached Iteration 5, System Testing and Refinement. All essential modules have been successfully developed and integrated into a single web-based application. These include account management, dataset uploading in various formats (CSV, Excel, JSON, and table images processed through OCR), automated rule-based data cleaning, dataset editing, descriptive statistics and data grouping, trend analysis, outlier detection, forecasting, interactive data visualizations, AI-assisted explanations, version history, and data export. At this stage, development efforts are primarily directed toward system testing, validating the accuracy of statistical results, and implementing improvements based on the panel's feedback and testing outcomes. These activities are being carried out to prepare the system for the upcoming pre-deployment defense.

Accomplishments Since the Last Report (Outline Presentation)

During the outline presentation, the panel gave ten (10) comments on the system and six (6) on the documentation. Table 1 shows how each was addressed. All ten system comments are now implemented; the diagrams were revised and accepted in the Outline Acceptance Form.

Panel comment	Status	How it was addressed
Limit the type of data the system can accept	Completed	Only CSV, Excel (.xlsx/.xls), JSON, PNG and JPG are accepted; other files are rejected with a clear message.
Set boundaries for the visualization tool	Completed	Charts are capped for readability: trend charts sample up to 250 points, group results show the top 50, and the correlation heatmap shows the 20 most related columns.
Use a progress bar when loading data	Completed	Uploads show a real progress bar; image (OCR) extraction shows a progress indicator.
Support adding columns	Completed	Columns can be added from a formula in Prepare or typed in directly in the Edit tab.
Allow renaming of uploads and file management	Completed	Datasets can be renamed, deleted and listed; every change is kept in a version history that can be restored.
Set the minimum file upload	Completed	Empty, header-only, and all-blank files are rejected; small but real datasets are still accepted.
Support downloading of results	Completed	Cleaned data and results can be exported as CSV, Excel, JSON and Parquet.
Support multiple users	Completed	Registration and login with expiring sessions; every request, including the AI features, checks that the user is logged in and owns the dataset.

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Panel comment	Status	How it was addressed
Suggest data from related fields (e.g., City "Panabo" → Zip Code "8105")	Completed	"Smart fill" finds such relationships, shows its match percentage and the rows it will fill, and is pre-selected when available; nothing changes until the user approves.
Support draft editing	Completed	The Edit tab keeps unsaved edits as a draft in the browser, with undo/redo and a Restore/Discard choice after a refresh.
Compare three LLMs in the documentation	Completed	Ollama, Groq and Gemini are compared in the manuscript (Table 1); Groq was selected.
Validation by a statistician	In progress	An internal numerical validation was completed (see Work Completed); a statistician has been contacted for the formal validation.
Revise the use case diagram	Completed	Revised and accepted in the Outline Acceptance Form
Change "decision support" to "AI insights"	Completed	Functional/non-functional requirements and the conceptual framework now use "AI insights".
Revise the CFD and DFD	Completed	Revised and accepted in the Outline Acceptance Form.
Improve the RRL	In progress	Being refined together with the final manuscript updates.

Table 1. Status of the panel's comments from the outline presentation

In addition to addressing the panel's feedback, the following improvements have been successfully implemented and tested:

- **Improved Smart Fill Reliability:** The Smart Fill feature has been enhanced to work effectively with real-world datasets, including those containing a high percentage of missing values, small amounts of data, and variations in spelling or capitalization (e.g., "Panabo", "panabo", and "PANABO "). It can also identify and reject misleading patterns, particularly in ID columns or columns where nearly all entries contain the same value.
- **Enhanced Data Integrity During Saving:** The system now immediately displays updated data after changes are saved. It also maintains the original arrangement of columns throughout the application, including exported files. Furthermore, numerical and date values retain their appropriate data types after editing.
- **Preservation of Leading Zeros in Codes:** The system has been improved to preserve leading zeros in identification codes and zip codes, such as "08100", during dataset uploads. These values are also correctly recognized as codes rather than numerical data, preventing their inclusion in statistical calculations, trend analysis, and outlier detection.
- **Date-Based Forecasting Improvements:** The Predict tab now supports the use of date columns when generating forecasts. It arranges records according to their actual dates and provides appropriate date labels for each predicted value, such as forecasts covering the next three months.

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- **Improved Interface for Non-Technical Users:** The user interface has been refined to make the system easier to understand and navigate. These enhancements include simpler and clearer labels, accurate descriptions of analytical methods such as "Outlier Detection (IQR)" and "Trend Analysis (linear regression)", confirmation messages before unsaved edits are discarded, and screen-reader labels for items in the fix checklist to improve accessibility.

Challenges Encountered and Solutions Implemented

Difficulty encountered	Strategy implemented
Development started later than planned (April instead of February), and June–July were taken up by documentation and enrollment requirements (diagrams, acceptance form).	Iterations were overlapped instead of run strictly one after another, and work was prioritized by the panel's comments. All development iterations were completed by early June; Iteration 5 was extended.
Real-world "messy" data broke simple rules: zip codes lost their leading zeros, dates like 03/01/2024 were misread, and edited numbers were saved as text.	Added explicit rules for codes, dates and number columns, each backed by automated tests so that a fix in one place does not break another.
The relationship-based suggestion (City → Zip Code) either missed real patterns or suggested false ones.	Redesigned the confidence measure: it now checks how often known values agree within each group and uses Goodman–Kruskal's lambda to confirm that the related column really improves the guess.
Saving results sometimes showed outdated data or changed the column order.	Fixed how saved versions are read back and refreshed, and tested the full save → view → export cycle.
Statistical results must be trustworthy for the statistician's validation.	Built a validation pack that recomputes every statistic with independent libraries (NumPy, SciPy, statsmodels) and compares the values.
OCR accuracy depends on image quality.	OCR output is shown for review and can be corrected in the Edit tab; this is stated as a limitation of the study.
Sending raw datasets to an online LLM would raise privacy concerns.	The AI receives only computed summaries (issues, counts, statistics), never the raw rows.
The Llama model originally planned on Groq was retired by the provider, so AI requests would have failed.	Compared the models still offered by Groq on the same Smartalyze prompt and selected openai/gpt-oss-120b for the clearest answers (about 2 seconds per reply); the model is a configuration setting, so it can be changed again without code changes.

Table 2. Challenges encountered and strategies implemented

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Comparison of Actual Progress to the Planned Timeline

Iteration	Planned (original schedule)	Actual	Status
1. Core System Setup	Feb 1 – Feb 14, 2026	Apr 16 – Apr 19, 2026: project setup, upload, version history, export, delete	Completed
2. Data Cleaning Module	Feb 14 – Mar 15, 2026	Apr 18 – May 31, 2026: rule-based cleaning, data-type fixes, smart fill, health score	Completed
3. Analysis & ML Integration	Mar 15 – Apr 13, 2026	May 2 – Jun 1, 2026: trend analysis, IQR outliers, AI explanations, forecasting	Completed
4. Visualization & Reporting	Apr 13 – May 16, 2026	May 26 – Jun 2, 2026: dashboard, overview, charts, export options	Completed
5. Testing & Refinement	May 16 – May 27, 2026	Aug 3, 2026 – present: panel revisions, automated tests, statistical validation, end-to-end testing	In progress

Table 3. Planned versus actual schedule (actual dates from the project's version-control history)

The development process started approximately ten weeks behind the original schedule, primarily due to documentation and enrollment requirements. Additionally, the system testing iteration was extended to accommodate the improvements recommended by the panel. Despite these delays, overlapping the development iterations allowed the team to complete all development activities under Iterations 1–4 by early June 2026. From April 16 to October 20, 2026, the project recorded a total of 123 commits. The remaining activities, including system validation, pilot testing, and deployment, are scheduled to take place before and after the pre-deployment defense, as detailed in the Work in Progress section.

Work Completed

Tasks Completed and Deliverables

Data ingestion: Supports uploading datasets in CSV, Excel, and JSON formats, as well as PNG/JPG images containing tables through Optical Character Recognition (OCR). The system also includes file-type validation, empty-file detection, and an upload progress bar to help users monitor the uploading process.

Automated data cleaning: Identifies missing values, duplicate records, inconsistent data types and date formats, hidden blank entries (e.g., "N/A"), variations in spelling, and outliers. The system also provides a list of recommended corrections for users to review and offers a Smart Fill feature that uses information from related columns to complete missing data.

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Dataset editing: Allows users to modify individual cells, add, rename, or remove columns and rows, reverse or restore changes using the undo and redo functions, and save their work as a draft for future editing.

Analysis: Provides descriptive statistical analysis, data grouping and summarization, correlation analysis, trend analysis using linear regression, outlier detection based on the Interquartile Range (IQR) method, and forecasting with an estimated range of values.

Visualization: Features a dashboard that displays the dataset health score, along with visual representations of trends and forecasts, bar charts for grouped data, data distribution charts, and a correlation heatmap.

AI-assisted explanations: Provides easy-to-understand data cleaning plans, clear explanations of analysis results, and an interactive follow-up chat feature called Ask AI. The system currently uses Groq as its configured and operational AI provider, running the openai/gpt-oss-120b model. Ollama and Gemini are also supported as alternative providers through the same interface. To maintain data privacy, only computed summaries are shared with the AI, while the original raw data rows are never transmitted.

Export and versioning: Allows users to export datasets in CSV, Excel, JSON, and Parquet formats. The system also maintains a version history by automatically saving each modification as a new version, enabling users to restore previous versions whenever necessary.

Security: Implements security measures such as password hashing, login sessions that automatically expire, and ownership verification for every dataset and AI request to ensure secure and authorized access.

Documentation: Updated the use case diagram, Data Flow Diagram (DFD), and CFD, which were subsequently accepted, along with the completion of the Large Language Model (LLM) comparison and the Outline Acceptance Form.

Evidence of Work

Code snippet 1. Smart fill only suggests a relationship that is consistent and truly informative (backend, pattern imputation service):

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```
weighted_confidence = round(measured_consistent / measured_support, 4)
if weighted_confidence < _MIN_CONFIDENCE:          # at least 80% of known values agree
    return None

# Goodman-Kruskal lambda: does knowing the City improve the guess
# compared with always using the single most common Zip Code?
global_hits = int(pd.concat(measured_known).value_counts().iloc[0])
lam = (measured_consistent - global_hits) / (measured_support - global_hits)
if lam < _MIN_LAMBDA:                              # must remove at least half the errors
    return None
```

Code snippet 2. Codes such as zip codes are kept as text; dates are not mistaken for codes (backend, snapshot service):

```
def has_leading_zero_codes(values: pd.Series) -> bool:
    # "08100" is a zip code (keep the zero); "03/01/2024" is a date, not a code.
    text = [v.strip() for v in values.dropna() if isinstance(v, str)]
    return any(re.fullmatch(r"[+-]?0\d+", v) for v in text)
```

Code snippet 3. Every AI request requires a valid login, and the AI receives only a computed summary (backend, AI routes):

```
@router.post("/suggest", response_model=AISuggestResponse)
async def ai_suggest(payload: AISuggestRequest,
                     authorization: str | None = Header(default=None),
                     db: AsyncSession = Depends(get_db)):
    await get_user_by_token(db, _require_token(authorization))   # 401 if not logged in
    suggestion = await get_suggestion(payload.context)          # summary only, no raw rows
```

Test results. Automated tests and end-to-end browser tests were run on the current version:

Test	What it checks	Result
Quality detection and cleaning tests	Spelling variants, disguised blanks, date formats, outliers, cleaning operations, smart fill, code columns	28 of 29 passed*
Snapshot and upload tests	Column order, number/date types after editing, leading zeros on upload, export	20 of 20 passed
Date-based forecasting tests	Chronological sorting, monthly steps, date labels, unreadable dates	4 of 4 passed
Statistical validation pack	42 values recomputed with NumPy, SciPy and statsmodels	41 of 42 matched**
End-to-end browser testing	Upload, clean, save, explore, detect, predict, edit and export on all six workspace tabs	Passed
AI explanations (Groq)	All four AI features (cleaning plan, cleaning chat, analysis explanation, analysis chat) through the system and in the browser; login required	Passed (about 2–4 s per answer)

Table 4. Summary of test results

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** One older test on spelling-variant grouping is still failing and is under review. ** The one difference is documentation, not a calculation error: the forecast range is computed as ±1 standard error, while the documentation states ±1.96; this will be settled with the statistician.*

Sample of the statistical validation (values from Smartalyze compared with independent libraries):

Statistic	Smartalyze	Independent result	Library	Match
Mean	176.7723	176.77228372	numpy.mean	Yes
Median	175.0754	175.07536902	numpy.median	Yes
Standard deviation (sample)	49.2095	49.20952424	numpy.std (ddof=1)	Yes
IQR upper fence	331.5386	331.53861357	numpy.percentile	Yes
Pearson correlation	0.8784	0.87844694	scipy.stats.pearsonr	Yes
Trend slope	2.590266	2.5902657790	scipy.stats.linregress	Yes
Forecast, step 1	1765.8631	1765.86309496	statsmodels OLS	Yes

Table 5. Excerpt of the statistical validation results

Work in Progress

Current task	Description	Anticipated completion
Formal statistical validation	Review of the statistical methods and validation results by a statistician; sign-off for the manuscript	November 6, 2026
Forecast range correction	Adjust the forecast range to the interval recommended by the statistician (e.g., a 95% prediction interval)	November 6, 2026
Pilot testing	Selected users try the system and give feedback (Implementation Plan, Phases 4–5)	October 26 – November 6, 2026
Remaining refinements	AI answer quality: complete answers (no cut-offs), clean formatting, use of Smartalyze's actual feature names, and accurate statistical wording. Also: suggested fixes in the Edit tab, time-zone display in History, file-size display, and plain-language type labels	October 31, 2026
Manuscript updates	Align the manuscript with the implemented system: remove "predictive analysis" from the Scope limitations (forecasting is implemented); mark draft saving as implemented (FR 4); describe passwords as hashed, not encrypted; correct the data dictionary (Users, AuthSessions); fill Table 9 with the actual test results; refine the RRL	November 9, 2026

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Current task	Description	Anticipated completion
Pre-deployment defense documents	Recommendation letter, evaluation sheet, and minutes of the outline presentation	November 9, 2026
Pre-deployment defense	Informal defense / formal defense	November 11–13 / 16–17, 2026
Final deployment	Deployment of the completed system (Implementation Plan, Phase 7)	After the formal defense

Table 6. Current tasks and anticipated completion dates

Conclusion

Summary of Key Findings and Achievements

Smartalyze is functionally complete. All ten system comments from the outline presentation have been implemented, the revised diagrams were accepted, and the system's statistical calculations were confirmed against independent libraries. Testing with realistic, messy data led to important improvements, especially in how the system handles codes, dates, saved versions and relationship-based suggestions. These improvements are protected by automated tests.

Assessment of Project Progress Against Goals

Objective (from the manuscript)	Status
Web-based data ingestion for CSV, Excel, JSON and table images (PNG, JPG)	Achieved (OCR accuracy depends on image quality)
Evaluate three LLMs and implement Groq for AI-assisted explanations	Achieved
Automated rule-based cleaning: missing values, duplicates, data types	Achieved
Recommendations for structuring, grouping, summarization and trends	Achieved
Interactive visualization dashboard	Achieved
Export of cleaned and processed data (CSV, Excel, JSON)	Achieved (Parquet also supported)

Table 7. Progress against the objectives of the study

Although development started later than planned, all development objectives have been met. What remains is the validation and testing work that confirms the system is accurate and usable, followed by deployment.

Plans and Next Steps

- Complete the formal validation by the statistician and apply the recommended forecast-range correction.

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- Conduct pilot testing with selected users and apply their feedback.
- Finish the remaining refinements and update the manuscript with the testing results.
- Prepare the pre-deployment defense requirements and present on November 11–17, 2026.
- Deploy the system after the formal defense, according to the System Implementation Plan.